

network, the MTSO would assign a different frequency for each of these callers and connect their calls. In a TDMA network, however, the MTSO divides a time frame (20 milliseconds in the first TDMA networks) into three time-slots, each 6.67 milliseconds long. Caller 1 will be assigned the first time-slot, caller 2 the second, and caller 3 the third. So for the first 6.67 milliseconds, the network handles caller 1's conversation, for the next 6.67 it handles caller 2's conversation, and so on. This cycle repeats every 20 milliseconds. Since the time between two time-slots is so small, neither party in the conversation realizes that they are only really speaking for a third of the time they are connected. The advantage is that all this is taking place on the *same* frequency band.

Let us now see what happens when TDMA and FDMA are combined. Suppose that our FDMA network has 200 frequency bands. This means that only 200 callers in each cell can make calls. Now if we employed TDMA for each frequency band, three callers could be handled per band. This means that by adding TDMA to our existing FDMA network, we have now brought up the capacity of the network to 600 callers per cell!

The *Telecommunication Industry Association (TIA)* defined the standards for TDMA networks. The Interim Standard 54 (IS-54) operates in the 800 MHz band, and the Interim Standard 136 (IS-136) operates in the 1900 MHz band. In both standards, the time frame of 20 milliseconds is divided into three time slots of 6.6 milliseconds each. In these time slots, only 3.3 milliseconds are actually used to transmit data; the other 3.3 serve as a 'guard space' between calls to prevent conversations from interfering with each other. In the USA, cellular service providers used the AMPS frequencies to implement their new TDMA technology, called Digital AMPS.

While going digital has many advantages, there are drawbacks with TDMA networks. The biggest of these is loss of voice quality. On an FDMA network, the voice signal, being analogue, sounds far better than it does on a digitised TDMA network. You've probably